

MA 251: Abstract Algebra

Course Code	MA 251
Course Name	Abstract Algebra
L-T-P-C	3-0-0-3
Intended for	B.Tech.
Distribution	Discipline Elective for B.Tech. Mathematics and Computing
Mutual Exclusion	MA 548
Approval	59th BoA (Apr 2025)

Course Contents

- **Unit I:** Binary operation and its properties, Groups: examples and basic properties, Subgroups, Finite Groups and Subgroups. [6 Lectures]
- **Unit II:** Cyclic groups, Order of a group, Permutation groups, Isomorphism theorems, Cayley's theorems. [6 Lectures]
- **Unit III:** Coset of a subgroup, Lagrange's theorem and its consequences, Normal subgroups, Quotient group, Internal direct products. [6 Lectures]
- **Unit IV:** Homomorphisms, Kernel Image of a homomorphism, Direct product of groups, Structure of finite abelian groups. [6 Lectures]
- **Unit V:** Rings: definition, examples and basic properties, Subrings, Zero divisors, Integral domains, Fields, Characteristic of a ring, Ideals, Prime ideal, Maximal ideals, Quotient rings. (6 Lectures)
- **Unit VI:** Ring Homomorphisms, Isomorphism theorems, Field of Quotients, Division Algorithm in Polynomial Rings and its consequences. [6 Lectures]
- **Unit VII:** Reducibility and Irreducibility tests, Unique factorization in $\mathbb{Z}[x]$, Irreducibles, Primes, UFD, PID and Euclidean domains. [6 Lectures]

Text Books

1. Joseph Gallian, **Contemporary Abstract Algebra**, 7th Edition.
2. N. Herstein, **Topics in Algebra**, Wiley Eastern Ltd., 1975.

References

1. M. Artin, **Algebra**, Prentice Hall of India, 1994.
2. J. B. Fraleigh, **A first Course in Abstract Algebra**, Narosa Publishing House, 2003.
3. Klima, Sigmon and Stitzinger, **Applications of Abstract Algebra with Maple and MATLAB**, 2nd Edition, 2006.